



CHARUSAT
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology & Engineering
Bachelor of Technology Programme
Information Technology
(B.Tech. IT)

ACADEMIC REGULATIONS & SYLLABUS

(Choice Based Credit System)



Faculty of Technology and Engineering
B. Tech. Programme
(Information Technology)
Chandubhai S. Patel Institute of Technology
Devang Patel Institute of Advance Technology and Research





CHARUSAT

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY AND ENGINEERING

ACADEMIC REGULATIONS Bachelor of Technology Programmes Choice Based Credit System

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. *System of Education*

Choice based Credit System with Semester pattern of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a course works in the chosen subject of specialization and also complete a project/dissertation if any. Apart from the Programme Core courses, provision for choosing University level electives and Programme/Institutional level electives are available under the Choice based credit system.

2. *Duration of Programme*

(i)	Undergraduate programme	(B. Tech.)
	Minimum	8 semesters (4 academic years)
	Maximum	16 semesters (8 academic years)

3. *Eligibility for admissions*

As enacted by Govt. of Gujarat from time to time.

4. *Mode of admissions*

As enacted by Govt. of Gujarat from time to time.

5. *Programme structure and Credits*

As per annexure – I attached

6. *Attendance*

6.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

6.2 Student attendance in a course should be Minimum 80%.

7 Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

- 7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and
- 7.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 70% of the marks for the course.

7.2 University Examination

- 7.2.1 The final examination by the University for 70% of the evaluation for the course will be through written paper and 100% for practical test or oral test or presentation by the student or a combination of any two or more of these.
- 7.2.2 In order to earn the credit in a course a student has to obtain grade other than FF.

7.3 Performance at Internal & University Examination

- 7.3.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations (internal/external) are as follows

Minimum marks in University Exam per subject	Minimum marks Overall per subject
40%	45%

- 7.3.2 A student failing to score 45% of the final examination will get a FF grade.
- 7.3.3 If a candidate obtains minimum required marks per subject but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8 Grading

8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Table: Grading Scheme (UG)

Range of Marks (%)	≥80	≥73 <80	≥66 <73	≥60 <66	≥55 <60	≥50 <55	≥45 <50	<45
Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

- (i) $SGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses in the semester
- (ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.
- (iii) No student will be allowed to move further if CGPA is less than 3 at the end of every academic year.
- (iv) A student will not be allowed to move to third year if he/she has not cleared all the courses of first year.

- (v) A student will not be allowed to move to fourth year if he/she has not cleared all the courses of second year.

9. Awards of Degree

9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

- 9.1.1 He should have earned at least minimum required credits as prescribed in course structure; and
- 9.1.2 He should have cleared all internal and external evaluation components in every course; and
- 9.1.3 He should have secured a minimum CGPA of 4.5 at the end of the programme;
- 9.1.4 In addition to above, the student has to complete the required formalities as per the regulatory bodies, if any.

9.2 The student who fails to satisfy minimum requirement of CGPA at the end of program will be allowed to improve the grades so as to secure a minimum CGPA for award of degree. Only latest grade will be considered.

10. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction:	CGPA $\geq$ 7.5
First class:	CGPA $\geq$ 6.0
Second Class:	CGPA $\geq$ 5.0

11. Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.



CHARUSAT
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**CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
(CHARUSAT)**

**FACULTY OF TECHNOLOGY & ENGINEERING
(FTE)**

CHOICE BASED CREDIT SYSTEM

FOR

BACHELOR OF TECHNOLOGY & ENGINEERING

Choice Based Credit System

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

1.1. Core Courses

1.1.1 University Core (UC)

University Core Courses are those courses which all students of the University of a Particular Level (PG/UG) will study irrespective of their Programme/specialisation.

1.1.2 Programme Core (PC)

A 'Core Course' is a course which acts as a fundamental or conceptual base for Chosen Specialisation of Engineering. It is mandatory for all students of a particular Programme and will not have any other choice for the same.

1.2 Elective Course (EC)

An 'Elective Course' is a course in which options / choices for course will be offered. It can either be for a Functional Course / Area or Streams of Specialization / Concentration which is / are offered or decided or declared by the University/Institute/Department (as the case may be) from time to time.

1.2.1 Institute Elective Course (IE)

Institute Courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialisation

1.2.2 Programme Elective Course (PE):

A 'Programme Elective Course' is a course for the specific programme in which students will opt for specific course(s) from the given set of functional course/ Area or Streams of Specialization options as offered or decided by the department from time-to-time

1.2.3 Cluster Elective Course (CE):

An 'Elective Course' is a course which students can choose from the given set of functional course/ Area or Streams of Specialization options (eg. Common Courses to EC/CE/IT/EE) as offered or decided by the Institute from time-to-time.

1.3 Non-Credit Course (NC) - AUDIT Course

A 'Non-Credit Course' is a course where students will receive Participation or Course Completion certificate. This will not be reflected in Student's Grade Sheet. Attendance and Course Assessment is compulsory for Non-Credit Courses

Charotar University of Science & Technology
Chandubhai S Patel Institute of Technology
Devang Patel Institute of Advance Technology and Research

Department of Information Technology

Vision

To become a front-runner for quality education, development and research in the field of IT.

Mission

- *To prepare next-generation technocrats for societal upliftment.*
- *To inculcate moral and ethical values for building vibrant nation.*

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)											
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN IT											
Sem	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours		Credit		Theory		Practical		Total
			Theory	Practical	Theory	Practical	Internal	External	Internal	External	
Final Year Sem-7	OCXXXXX	Elective 7.1	3	2	3	1	30	70	25	25	150
	OCXXXXX	Elective 7.2	3	2	3	1	30	70	25	25	150
	IT452	Minor Project	0	18	0	9	0	0	125	175	300
	IT446	Summer Internship-II	0	3	0	3	0	0	75	75	150
			6	25	6	14	50	150	250	250	750
Final Year Sem-8	IT461	Software Project Major	0	36	0	18	0	0	250	350	600
			0	36	0	18	0	0	250	350	600

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)				
LIST OF ELECTIVE SUBJECTS FOR B TECH PROGRAMME IN IT				
ELECTIVES	Code	Elective – 7.1	Code	Elective – 7.2
	OCCSE4001	Reinforcement Learning	OCCSE4002	Social Network Analysis
	OCCE4001	Software Testing	OCCE4002	Introduction to Industry 4.0 and Industrial Internet of Things
	OCIT4001	Privacy and Security in Online Social Media	OCIT4002	Practical Cyber Security for Cyber Security Practitioners
	OCAIML4001	Responsible & Safe AI System	OCAIML4002	Introduction to Large Language Models

B. Tech. (Information Technology) Programme

SYLLABI (Semester – 7)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
OCCSE4001: Reinforcement Learning (Elective – 7.1)

Description:

This course OCCSE4001: Reinforcement Learning is offered from SWAYAM as noc25_cs62 – Reinforcement Learning.

URL: https://onlinecourses.nptel.ac.in/noc25_cs62/preview

A. Credits and Week:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

* The practical component is offered by CHARUSAT

B. About this course:

Reinforcement learning is a paradigm that aims to model the trial-and-error learning process that is needed in many problem situations where explicit instructive signals are not available. It has roots in operations research, behavioral psychology and AI. The goal of the course is to introduce the basic mathematical foundations of reinforcement learning, as well as highlight some of the recent directions of research.

C. Course Layout:

Week 1: Introduction to Reinforcement Learning (RL): RL Framework and Application, Introduction to immediate RL, Bandit Optimalities, Value Function Based Methods.

Week 2: Bandit Algorithms: Upper Confidence Bound (UCB), Probably Approximately Correct (PAC) Bounds, Median Elimination, Thompson Sampling.

Week 3: Bandit Algorithms: Policy Search, REINFORCE, Contextual Bandits, Full RL introduction, Returns, Value Functions and Markov Decision Processes (MDPs).

Week 4: Markov Decision Processes (MDPs): MDP Modelling, Bellman Equation, Bellman Optimality Equation, Cauchy Sequence and Green's Equation, Banach Fixed Point Theorem, Convergence Proof.

Week 5: Bellman Optimality: Lpi Convergence, Value Iteration, Policy Iteration, Dynamic Programming, Monte Carlo (MC), Control in Monte Carlo.

Week 6: Dynamic Programming & Temporal Difference (TD) Methods: Off Policy MC, Upper Confidence bounds applied to Trees, TD (0) Control, Q-Learning, Afterstate.

Week 7: Eligibility Traces: Eligibility Traces, Backward view of Eligibility Traces, Eligibility Trace Control, Thompson Sampling Recap.

Week 8: Function Approximation: Linear Parameterization, State Aggregation Methods, Function Approximation and Eligibility Traces, Least Squares Temporal Difference, Least Squares Temporal Difference for Q-function, and Least Squares Policy Iteration.

Week 9: Least Squares Methods: Deep Q-Network and Filtered Q- Iteration, Policy Gradient Approach, Actor Critic, Policy Gradient with Function Approximation.

Week 10: Fitted Q, DQN & Policy Gradient for Full RL: Hierarchical Reinforcement Learning, Types of Optimality, Semi Markov Decision Process, Options, Learning with Options, Hierarchical Abstract Machines.

Week 11: Hierarchical RL: MAXQ, MAXQ Value Function Decomposition, Option Discovery.

Week 12: Partially Observable Markov Decision Processes (POMDPs): Introduction to POMDP, Solving POMDP.

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the foundational concepts of Reinforcement Learning, including the agent-environment interaction framework, and analyze simple decision-making problems using Multi-Armed Bandit algorithms.
CO2	Apply Markov Decision Processes (MDPs) and Bellman equations to model sequential decision-making tasks and evaluate optimal policies using Dynamic Programming.
CO3	Analyze Temporal Difference methods and eligibility traces to enhance learning efficiency in value function estimation.
CO4	Implement function approximation techniques and Least Squares methods to enable value estimation in large or continuous state spaces.
CO5	Design and develop advanced RL agents using Deep Q-Networks (DQN), Policy Gradients, Hierarchical RL, and POMDP frameworks to solve complex decision-making problems.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO9	PO1 0	PO1 1	PO12	PSO 1	PSO 2
CO1	3	2	-	-	-	2	-	2	-	1	-	2	3	1
CO2	3	3	3	2	3	1	-	-	-	-	-	1	3	2
CO3	3	3	2	3	3	-	1	-	2	2	2	2	2	3
CO4	3	2	2	2	3	2	2	-	1	-	1	2	2	2
CO5	3	2	3	3	2	3	2	-	2	3	2	3	3	3

The correlation levels 1, 2, or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, “-” is placed.

D. Books and references

1. R. S. Sutton and A. G. Barto. Reinforcement Learning - An Introduction. MIT Press. 1998.
2. Maxim Lapan. Practical Deep Reinforcement Learning, June 21, 2018.
<https://www.goodreads.com/book/show/37773354-deep-reinforcement-learning-hands-on>
3. Research Papers

E. Criteria To Get a Certificate

- Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.
- Exam score = 75% of the proctored certification exam score out of 100 Final score = Average assignment score + Exam score
- YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$. If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.
- Certificate will have your name, photograph and the score in the final exam with the breakup. It will have the logos of NPTEL and IIT Madras. It will be e-verifiable at nptel.ac.in/noc.

Description:

This course OCCE4001– Software Testing is offered from SWAYAM as noc24_cs91 – Software Testing.

URL: https://onlinecourses.nptel.ac.in/noc24_cs91/preview

A. Credits and Week:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

The practical component is offered by CHARUSAT

B. About this course:

This course will cover various techniques for test case design, as used for testing of software artifacts including requirements, design and code. We will discuss algorithms and techniques for test case design based on graphs, logic, syntax of programming languages and on inputs. Special techniques for testing object-oriented features and web applications will also be discussed. The course will end with symbolic testing techniques. These broadly will cover test cases for both white-box and black-box.

C. Course Layout:

Week 1: Techniques and algorithms for test case design: Graph-based testing- structural coverage criteria.

Week 2: Graphs based testing: Data flow coverage criteria

Week 3: Graphs based testing: Data flow coverage criteria

Week 4: Graphs coverage for source code, design elements, and requirements

Week 5: Techniques and algorithms for test case design: Logic-based testing- Predicates, logic-based coverage criteria

Week 6: Specification based logic coverage, logic coverage on finite state machines

Week 7: Input space partitioning: Input domain modeling, combination strategies criteria

Week 8: Syntax based testing: Coverage criteria based on syntax, mutation testing

Week 9: Test case design (as learnt above) applied to object-oriented applications

Week 10: Test case design (as learnt above) applied to web applications

Week 11: Symbolic testing

Week 12: Concolic testing, Conclusion

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Apply structural and data flow coverage criteria for test case design using graph - based techniques.
CO2	Develop and evaluate logic-based and specification-based test cases, including finite state machine-based testing.
CO3	Design effective test cases using input space partitioning and syntax-based testing techniques, including mutation testing.
CO4	Implement test case design strategies for object-oriented and web-based applications based on learned techniques.
CO5	Analyze and apply advanced testing techniques such as symbolic and concolic testing to enhance test effectiveness.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	2	2	-	-	-	-	-	-	-	3	2
CO2	3	3	-	3	2	-	-	-	-	-	-	-	3	2
CO3	3	2	-	3	2	-	-	-	-	-	-	-	3	2
CO4	2	-	2	3	3	-	-	-	-	-	-	-	3	3
CO5	3	2	3	3	3	-	-	-	-	-	-	-	3	3

The correlation levels 1, 2, or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High).

If there is no correlation, “-” is placed.

D. Books and References:

NIL

E. Criteria To Get a Certificate

- Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.
- Exam score = 75% of the proctored certification exam score out of 100 Final score = Average assignment score + Exam score
- YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE

ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$. If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.

- Certificate will have your name, photograph and the score in the final exam with the breakup. It will have the logos of NPTEL and IIIT Bangalore. It will be e-verifiable at nptel.ac.in/noc.

OCIT4001: Privacy and Security in Online Social Media (Elective – 7.1)

Description:

This course, OCIT4001: Privacy and Security in Online Social Media, is offered by SWAYAM as noc25_cs79.

URL: https://onlinecourses.nptel.ac.in/noc25_cs79/preview

A. Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

* Practical component is offered by CHARUSAT

B. About this course:

With increase in the usage of the Internet, there has been an exponential increase in the use of online social media and networks on the Internet. Websites like Facebook, YouTube, LinkedIn, Twitter, Flickr, Instagram, Google+, Foursquare, Pinterest, Tinder, and the likes have changed the way the Internet is being used. However, widely used, there is a lack of understanding of privacy and security issues on online social media. Privacy and security of online social media need to be investigated, studied and characterized from various perspectives (computational, cultural, psychological, etc.). Student completing the course will be able to appreciate various privacy and security concerns (spam, phishing, fraud nodes, identity theft) on Online social media and Student will be able to clearly articulate one or two concerns comprehensively on one Online social media, this will be achieved by homework.

PREREQUISITES: Basic / Intermediate programming course. Understanding of Python will be necessary for the course. Should be able to quickly learn APIs, and to collect data from social networks.

INDUSTRY SUPPORT: Any company which is interested in social media/networks data will be interested in recruiting the students finishing the course.

C. Course Layout:

Week 1: What is Online Social Networks, data collection from social networks, challenges, opportunities, and pitfalls in online social networks, APIs

Week 2: Collecting data from Online Social Media.

Week 3: Trust, credibility, and reputations in social systems

Week 4: Trust, credibility, and reputations in social systems

Week 5: Online social Media and Policing

Week 6: Information privacy disclosure, revelation and its effects in OSM and online social networks

Week 7: Phishing in OSM & Identifying fraudulent entities in online social networks

Week 8: Refresher for all topics

Week 9 to 12: Research paper discussion

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the structure, functionality, and privacy/security issues of various Online social media (OSM) platforms.
CO2	Analyze and apply modern tools and techniques to detect security threats such as phishing, fraud, and identity theft in online social networks using programming and data analysis skills.
CO3	Demonstrate the ability to critically evaluate real-world case studies and research literature to propose privacy-preserving strategies and ethically responsible practices in social media ecosystems.

Course Articulation Matrix:

	P O1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	2	-	-	3	2	-	2	-	2	-	2	3	2
CO2	3	3	2	2	3	-	-	2	-	-	-	2	3	3
CO3	2	3	2	3	2	2	2	3	2	3	2	3	2	3

The correlation levels 1, 2, or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, “-” is placed.

D. Books and References

1. Official Material <https://nptel.ac.in/courses/106106146>

E. Criteria To Get a Certificate

- Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.
- Exam score = 75% of the proctored certification exam score out of 100 Final score = Average assignment score + Exam score
- Please note that assignments encompass all types (including quizzes, programming tasks, and essay submissions) available in the specific week.
- YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$. If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.
- Certificate will have your name, photograph and the score in the final exam with the breakup. It will have the logos of NPTEL and IIT Delhi. It will be e-verifiable at nptel.ac.in/noc.

OCAIML4001: Responsible & Safe AI Systems (Elective – 7.1)

Description:

This course OCAIML4001: Responsible & Safe AI Systems is offered from SWAYAM as noc24_cs132.

URL: https://onlinecourses.nptel.ac.in/noc24_cs132/preview

A. Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

The practical component is offered by CHARUSAT.

B. About this course:

This course provides students with a comprehensive understanding of the ethical, social, and safety considerations essential for developing and deploying artificial intelligence (AI) systems. Uncover the intricacies of algorithmic transparency, fairness in machine (un)learning, interpretability, consistency and many more. The course encourages critical thinking and fosters a deep appreciation for the impact of AI on individuals and communities. Students who complete the course can: recognize possible harms that can be caused by modern AI capabilities; learn to reason about various perspectives on the trajectory of AI development and proliferation; learn about latest research agendas towards making AI systems safer.

C. Course Layout:

Week 1 & 2: AI Capabilities Improvement in last 5-10 years , Imminent risks from AI Models:

Toxicity, bias, goal misspecification, adversarial examples etc. , Long-term risks from AI Models: Misuse, Misgeneralization, Rogue AGI , Principles of RAI - Transparency; Accountability; Safety, Robustness and Reliability; Privacy and Security; Fairness and non-discrimination; Human-Centred Values; Inclusive and Sustainable development, Interpretability, Recap of Deep Learning Techniques, Language/Vision Models , AI Risks for Gen models, Adversarial Attacks – Vision, NLP, Superhuman Go agents.

Week 3 & 4: ML Poisoning Attacks like Trojans, Implications for current and future AI safety, Explainability, Imminent and Long-term potential for transparency techniques, Mechanistic

Interpretability, Representation Engineering, model editing and probing, Critiques of Transparency for AI Safety

Week 5 & 6: Privacy & Fairness in AI

Week 7 & 8: Metrics and Tools for RAI - measuring bias/fairness, adversarial testing, explanations (Lime/SHAP/GradCam), audit mechanisms, Regulation landscape - DPDP act (India), GDPR (EU), EU AI act, US presidential declaration, Ethical approvals, informed consent, participatory design, future of work, Indian context, what is AGI? When could it be achieved? , Instrumental Convergence: Power Seeking, Deception etc.

Week 9 & 10: RAI in Legal domain, RAI in Health care domain, RAI in Education domain, A few other domains, Policy issues in RAI.

Week 11 & 12: Couple of panel discussion with industry practitioners, academic, government (possibly), and others, Fireside chat with eminent personalities, Recorded Paper reading discussion

Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Identify ethical, legal, and social issues related to the development and deployment of AI systems.
CO2	Analyze real-world AI applications to assess fairness, transparency, and potential biases.
CO3	Demonstrate understanding of algorithmic accountability, interpretability, and decision-making consistency.
CO4	Evaluate the safety and robustness of AI systems using contemporary tools, frameworks, and case studies.
CO5	Apply principles of responsible AI in the design and critique of machine learning models and data pipelines.
CO6	Interpret and discuss current research and policy directions in AI safety, governance, and regulation.
CO7	Communicate informed perspectives on the ethical implications and future trajectory of AI technologies.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO1 1	PO1 2
CO1	2	1	–	–	–	3	3	3	–	2	–	2
CO2	3	2	2	–	–	3	3	3	1	2	–	2
CO3	3	2	2	–	–	3	2	2	1	2	–	2

CO4	3	3	3	2	2	2	2	3	2	2	–	2
CO5	3	2	3	2	2	3	3	3	1	3	1	3
CO6	2	2	1	–	–	3	3	3	2	2	–	2
CO7	2	1	–	–	–	3	3	3	3	3	–	2

The correlation levels 1, 2, or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, “–” is placed.

D. Books and References:

NIL

E. Criteria To Get a Certificate

- Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.
- Exam score = 75% of the proctored certification exam score out of 100 Final score = Average assignment score + Exam score
- YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$. If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.
- Certificate will have your name, photograph and the score in the final exam with the breakup. It will have the logos of NPTEL and IIT Hyderabad .It will be e-verifiable at nptel.ac.in/noc.

OCCSE4002: Social Network Analysis (Elective – 7.2)

Description:

This course, OCCSE4002: Social Network Analysis, is offered by SWAYAM as noc24_cs90—Social Network Analysis.

URL: https://onlinecourses.nptel.ac.in/noc24_cs90/preview

A. Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

The practical component is offered by CHARUSAT.

B. About this course:

Networks are a fundamental tool for modeling complex social, technological, and biological systems. Coupled with the emergence of online social networks and large-scale data availability in social sciences, this course focuses on the analysis of massive networks which provide many computational, algorithmic, and modeling challenges. The course will cover research on the structure and analysis of such large networks and on models and algorithms that abstract their basic properties. We will explore how to practically analyze large-scale network data and how to reason about it through models for network structure and evolution. Topics covered in this course are how information spreads through society; robustness and fragility of networks; algorithms for the World Wide Web; prediction and recommendation in online social networks; representation learning for large networks; etc.

PREREQUISITES: Python programming, Probability and Statistics, Machine Learning.

INDUSTRY SUPPORT: Any social media company, E-commerce company, etc

C. Course Layout:

Week 1: Introduction ; Tutorial 1: Introduction to Python/Colab ; Tutorial 2: Introduction to NetworkX - Part I

Week 2: Network Measures ; Tutorial 3: Introduction to NetworkX - Part II

Week 3: Network Growth Models

Week 4: Link Analysis

Week 5: Tutorial 4: Graph Visualization Tools; Community Detection - Part I

Week 6: Community Detection - Part II

Week 7: Link Prediction

Week 8: Cascade Behavior and Network Effects

Week 9: Anomaly Detection

Week 10: Introduction to Deep Learning; Graph Representation Learning - Part I

Week 11: Graph Representation Learning - Part II; Tutorial: Coding on Graph Representation Learning

Week 12: Applications and Case Studies ; Conclusion

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the fundamentals of network analysis, including graph theory, key concepts such as nodes, edges, and network structures, and their relevance in real-world applications.
CO2	Apply Python and the NetworkX library to model, analyze, and visualize networks, demonstrating practical proficiency in handling graph-based data.
CO3	Analyze and implement network growth models and link analysis techniques, evaluating their impact on network structure and behavior.
CO4	Apply community detection methods and anomaly detection techniques to identify patterns and outliers in complex networks, and assess their effectiveness in real-world datasets.
CO5	Understand and implement graph representation learning techniques using deep learning models, and apply these methods to enhance performance in network- related tasks such as link prediction and node classification.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	3	2	1	2	2	1	1	1	2	1	1	2	1
CO2	3	2	3	2	3	1	2	1	2	1	2	2	1	2
CO3	3	3	3	2	2	1	2	2	2	2	1	1	3	3
CO4	2	2	3	3	2	3	2	2	2	1	2	2	2	2
CO5	3	3	2	2	3	2	3	3	3	3	3	3	3	3

The correlation levels 1, 2, or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, “-” is placed.

D. Books and References

1. Social Network Analysis, Tanmoy Chakraborty, Wiley, 2021
2. Network Science, Albert-Lazzlo Barabasi
3. Social Network Analysis: Methods and Applications, Stanley Wasserman, Katherine Faus

E. CRITERIA TO GET A CERTIFICATE

- Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.
- Exam score = 75% of the proctored certification exam score out of 100 Final score = Average assignment score + Exam score
- YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$. If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.
- Certificate will have your name, photograph and the score in the final exam with the breakup. It will have the logos of NPTEL and IIT Delhi. It will be e-verifiable at nptel.ac.in/noc.

OCCE4002: Introduction to Industry 4.0 and Industrial Internet of Things (Elective – 7.2)

Description:

This course OCCE4002: Introduction to Industry 4.0 and Industrial Internet of Things is offered from SWAYAM as noc25_cs43 – Introduction to Industry 4.0 and Industrial Internet of Things.

URL: https://onlinecourses.nptel.ac.in/noc25_cs43/preview

A. Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

* Practical component is offered by CHARUSAT

B. About this course:

Industry 4.0 concerns the transformation of industrial processes through the integration of modern technologies such as sensors, communication, and computational processing. Technologies such as Cyber Physical Systems (CPS), Internet of Things (IoT), Cloud Computing, Machine Learning, and Data Analytics are considered to be the different drivers necessary for the transformation. Industrial Internet of Things (IIoT) is an application of IoT in industries to modify the various existing industrial systems. IIoT links the automation system with enterprise, planning and product lifecycle. This course has been organized into the following modules.

C. Course Layout:

Week 1: Introduction: Sensing & actuation, Communication-Part I, Part II, Networking-Part I, Part II

Week 2: Industry 4.0: Globalization and Emerging Issues, The Fourth Revolution, LEAN Production Systems, Smart and Connected Business Perspective, Smart Factories

Week 3: Industry 4.0: Cyber Physical Systems and Next Generation Sensors, Collaborative Platform and Product Lifecycle Management, Augmented Reality and Virtual Reality, Artificial Intelligence, Big Data and Advanced Analysis

Week 4: Cybersecurity in Industry 4.0, Basics of Industrial IoT: Industrial Processes-Part I, Part II, Industrial Sensing & Actuation, Industrial Internet Systems.

Week 5: IIoT-Introduction, Industrial IoT: Business Model and Reference Architecture: IIoT- Business Models-Part I, Part II, IIoT Reference Architecture-Part I, Part II.

Week 6: Industrial IoT- Layers: IIoT Sensing-Part I, Part II, IIoT Processing-Part I, Part II, IIoT Communication-Part I.

Week 7: Industrial IoT- Layers: IIoT Communication-Part II, Part III, IIoT Networking-Part I, Part II, Part III.

Week 8: Industrial IoT: Big Data Analytics and Software Defined Networks: IIoT Analytics - Introduction, Machine Learning and Data Science - Part I, Part II, R and Julia Programming, Data Management with Hadoop.

Week 9: Industrial IoT: Big Data Analytics and Software Defined Networks: SDN in IIoT- Part I, Part II, Data Center Networks, Industrial IoT: Security and Fog Computing: Cloud Computing in IIoT-Part I, Part II.

Week 10: Industrial IoT: Security and Fog Computing - Fog Computing in IIoT, Security in IIoT- Part I, Part II, Industrial IoT- Application Domains: Factories and Assembly Line, Food Industry.

Week 11: Industrial IoT- Application Domains: Healthcare, Power Plants, Inventory Management & Quality Control, Plant Safety and Security (Including AR and VR safety applications), Facility Management.

Week 12: Industrial IoT- Application Domains: Oil, chemical and pharmaceutical industry, Applications of UAVs in Industries, Real case studies:

Case study - I: Milk Processing and Packaging Industries

Case study - II: Manufacturing Industries - Part I

Case study - III: Manufacturing Industries - Part II

Case study - IV: Student Projects - Part I

Case study - V: Student Projects - Part II

Case study - VI: Virtual Reality Lab

Case study - VII: Steel Technology Lab

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the fundamentals of Industry 4.0, including its evolution, drivers, and key technologies such as smart factories, CPS, and digital transformation.
CO2	Explain the structure and components of Industrial IoT systems, including sensing, actuation, communication, and layered architectures.
CO3	Evaluate communication protocols, networking models, and data processing methods used in IIoT, including big data tools and machine learning applications.

CO4	Assess security, privacy, and cloud/fog computing frameworks relevant to the implementation of secure IIoT systems.
Co5	Apply concepts of IIoT and Industry 4.0 to real-world industrial domains through case studies and analyze their effectiveness and impact.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	-	2	2	1	-	-	1	1	1	1	1
CO2	3	3	2	2	3	1	-	1	-	2	1	2	2	1
CO3	3	3	2	3	3	-	1	1	2	2	2	2	2	3
CO4	2	2	2	2	3	2	2	2	1	-	1	2	2	2
CO5	2	2	3	3	2	3	2	2	3	3	2	3	3	3

The correlation levels 1, 2, or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If there is no correlation, “-” is placed.

D. Books and references

1. S. Misra, A. Mukherjee, and A. Roy, 2020. Introduction to IoT. Cambridge University Press. Availability: https://www.amazon.in/Introduction-IoT-SudipMisra/dp/1108959741/ref=sr_1_1?dchild=1&keywords=sudip+misra&qid=1627359928&sr=8-1
2. S. Misra, C. Roy, and A. Mukherjee, 2020. Introduction to Industrial Internet of Things and Industry 4.0. CRC Press. Availability: https://www.amazon.in/dp/1032146753/ref=sr_1_3?dchild=1&keywords=sudip+misra&qid=1627359971&sr=8-3
3. Research Papers

E. Criteria To Get a Certificate

- Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.
- Exam score = 75% of the proctored certification exam score out of 100 Final score = Average assignment score + Exam score
- YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$. If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.
- Certificate will have your name, photograph and the score in the final exam with the breakup.

It will have the logos of NPTEL and IIT Kharagpur. It will be e-verifiable at nptel.ac.in/noc.

OCIT4002: Practical Cyber Security for Cyber Security Practitioners (Elective – 7.2)

Description:

This course, OCIT4002: Practical Cyber Security for Cyber Security Practitioners, is offered by SWAYAM as noc24_cs85.

URL: https://onlinecourses.nptel.ac.in/noc24_cs85/preview

A. Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

The practical component is offered by CHARUSAT.

B. About this course:

As we interact with various industry and their needs for cyber security, they often complain that the courses at academic institutions provide students with the fundamentals of cyber security, but the students are often not familiar with the practical issues associated with running a cyber-security operation in a real organization. Also, while a lot of the theses at master's level or dissertation at the PhD level are based on theoretical exploration of techniques, algorithms, or mathematical modelling of cyber security problems, there is a dearth of dissertation and thesis on the practical aspects of running cyber security operations. Having worked very closely with industry, we appreciate such concerns because there are very interesting problems to be solved for practical cyber security which also requires a great deal of fundamental thinking, reasoning, and concerns about human cognition of the inputs the operators and leaders receive from cyber security tools, dashboards, policy formulations but then in the crowd of details, fundamentals are lost, reasoning is replaced by ad hoc decisions. To change that, we need workforce, as well as academics who can abstract out the fundamentals from such practical problems and develop various practically oriented methods, methodologies, policy frameworks, and decision processes which would simplify cyber operations. This course is meant to be a first step towards that line of thinking, modeling, and methodology development.

PREREQUISITES: A previous course on fundamentals of Cyber Security is desirable, if not, courses in Operating System, Networking, Software Engineering, and Programming will be desirable.

C. Course Layout:

Week 1: Introduction: Cyber Security Posture, Maturity, and Continuous Improvement Strategies

Week 2: Knowing your attacker: MITRE ATT&CK Framework

Week 3: Knowing your attacker: MITRE ATT&CK Framework

Week 4: Cyber Kill Chains: Lockheed Martin Cyber Kill Chain and Unified Kill Chain

Week 5: Defending Yourself: MITRE DEF3ND Framework

Week 6: Securing your Configurations: STIG Benchmarking

Week 7: Sharing Threat Intelligence: STIX and TAXII 2.0

Week 8: Cyber Security Risk Management

Week 9: Cyber Resilience Assessment

Week 10: Cyber Security Policy Formulation

Week 11: Cyber Security Policy Formulation

Week 12: Cyber Crisis Management Plan and Cyber Incident Management Plan

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand and apply practical methodologies, tools, and frameworks used in real-world cyber security operations such as MITRE ATT&CK, DEF3ND, and Cyber Kill Chain.
CO2	Analyze and evaluate organizational cyber security postures through configuration benchmarking, threat intelligence sharing (e.g., STIX, TAXII), and risk management strategies.
CO3	Design, formulate and assess effective cyber security policies, resilience plans, and incident response strategies aligned with industrial best practices.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO4	PO5	PO6	PO 7	PO8	PO 9	PO1 0	PO11	PO12	PS O1	PSO 2
CO1	3	2	2	3	3	2	1	2	2	2	1	2	3	2
CO2	2	3	2	3	3	3	2	2	2	2	2	2	3	3
CO3	2	3	3	3	2	3	3	3	2	3	3	2	2	3

The correlation levels 1, 2, or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If there is no correlation, “-” is placed.

D. Books and References

1. Official Material <https://nptel.ac.in/courses/106104467>

E. Criteria To Get a Certificate:

- Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.
- Exam score = 75% of the proctored certification exam score out of 100 Final score = Average assignment score + Exam score
- YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$. If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.
- Certificate will have your name, photograph and the score in the final exam with the breakup. It will have the logos of NPTEL and IIT Kanpur. It will be e-verifiable at nptel.ac.in/noc.

OCAIML4002: Introduction to Large Language Models (Elective – 7.2)

Description

This course OCAIML4002: Introduction to Large Language Models (LLMs) is offered from SWAYAM as noc25_cs45.

URL: https://onlinecourses.nptel.ac.in/noc25_cs45/preview

A. Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

The practical component is offered by CHARUSAT.

B. About this course :

This course introduces the fundamental concepts underlying Large Language Models (LLMs). It starts with an introduction to the various problems in NLP, and discusses how to approach the problem of language modeling using deep learning. It describes the architectural intricacies of Transformers and the pre-training objectives of the different Transformer-based models. It also discusses the recent advances in LLM research, including LLM alignment, prompting, parameter-efficient adaptation, hallucination, bias and ethical considerations.

C. Course Layout:

Week 1: Course Introduction, Introduction to NLP (NLP Pipeline, Applications of NLP).

Week 2: Introduction to Statistical Language Models, Statistical Language Models: Advanced Smoothing and Evaluation.

Week 3: Introduction to Deep Learning (Perceptron, ANN, Backpropagation, CNN), Introduction to PyTorch.

Week 4: Word Representation, Word2Vec, fast Text, GloVe , Tokenization Strategies

Week 5: Neural Language Models, CNN, RNN, LSTM, GRU, Sequence-to-Sequence Models, Greedy Decoding, Beam search, Other Decoding Strategies: Nucleus Sampling, Temperature Sampling, Top-k Sampling, Attention in Sequence-to-Sequence Models.

Week 6: Introduction to Transformers, Self and Multi-Head Attention, Positional Encoding and Layer Normalization, Implementation of Transformers using PyTorch,

Week 7: Pre-Training Strategies: ELMo, BERT (Encoder-only Model), Pre-Training Strategies: Encoder-decoder and Decoder-only Models, Introduction to HuggingFace.

Week 8: Instruction Tuning, Prompt-based Learning, Advanced Prompting Techniques and Prompt Sensitivity, Alignment of Language Models with Human Feedback (RLHF).

Week 9: Open-book question answering: The case for retrieving from structured and unstructured sources; retrieval-augmented inference and generation, Retrieval augmentation techniques, Key-value memory networks in QA for simple paths in KGs, Early HotPotQA solvers, pointer networks, reading comprehension, REALM, RAG, FiD, Uniformed, KGQA (e.g., EmbedKGQA, GrailQA).

Week 10: Knowledge graphs (KGs), Representation, completion, Tasks: Alignment and isomorphism Distinction between graph neural networks and neural KG inference.

Week 11: Parameter-efficient Adaptation (Prompt Tuning, Prefix Tuning, LoRA), An Alternate Formulation of Transformers: Residual Stream Perspective, Interpretability Techniques.

Week 12: Overview of recently popular models such as GPT-4, Llama-3, Claude-3, Mistral, and Gemini, Ethical NLP – Bias and Toxicity, Conclusion.

Course Outcomes (COs):

At the end of the course, students will be able to

CO1	Understand and explain the fundamental problems and tasks in Natural Language Processing (NLP).
CO2	Apply deep learning techniques to solve language modeling tasks
CO3	Describe the architecture and components of Transformer models.
CO4	Analyze and compare pre-training objectives used in various Transformer- based language models.
CO5	Evaluate recent advancements in LLM research, including alignment, prompting techniques, bias, hallucination, and ethical considerations.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	-	-	-	-	-	-	2
CO2	3	3	2	2	3	-	-	-	-	-	-	2
CO3	3	2	-	2	3	-	-	-	-	-	-	2
CO4	3	3	2	3	3	-	-	-	-	-	-	2
CO5	2	2	2	2	2	-	-	3	-	-	-	3

D. Books and references

NIL

E. Criteria To Get a Certificate:

- Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.
- Exam score = 75% of the proctored certification exam score out of 100 Final score = Average assignment score + Exam score
- YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE $\geq 10/25$ AND EXAM SCORE $\geq 30/75$. If one of the 2 criteria is not met, you will not get the certificate even if the Final score $\geq 40/100$.
- Certificate will have your name, photograph and the score in the final exam with the breakup. It will have the logos of NPTEL and IIT Delhi .It will be e-verifiable at nptel.ac.in/noc.

IT452: MINOR PROJECT

A. Credits and Hours:

Teaching Scheme	Theory	Project	Total	Credit
Hours/week	0	18	18	9
Marks	0	300	300	

B. Outline of the Course:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work weekly to respective internal guide.
- Project will be evaluated at least once per week in laboratory during the semester and final submission at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of Software Engineering.
- Students have to submit project with following listed documents at the time of final submission.
- Project Synopsis
- Software Requirement Specification
- SPMP
- Final Project Report/paper
- Project Setup file with Source code [Uploaded on GitHub]
- Project Presentation (PPT)
- Video Recording (Per Project) A student has to produce some useful outcome by conducting experiments or project work.
- A student has to produce some useful outcome by conducting experiments or project work.

Total hours (Theory): 00
Total hours (Lab): 270
Total hours: 270

C. Instructional Method and Pedagogy:

1. Project Groups would be form of maximum two students.
2. Inter batch group formation is not permitted due to difficulties in progress tracking.
3. Students are advised to choose innovative and challenging definitions.
4. Batch wise project definitions must be unique.
5. Any management system would not be encouraged.
6. Tools like GitHub would be used to track the progress of project development by the concern faculty.
Concerned guide will demonstrate the working of GitHub Tool.
7. Student must prepare report at end of semester as part of submission.
8. Report structure is finalized for semester end submission.
9. To have a better outcome as well as progress tracking at the end of semester, it is decided that students must appear for two internal reviews, which will help them to get more insight in the project.
10. To maintain similarity below 40%, Students must submit project's final document to concern SGP guide for plagiarism check before 15 days of external exam.
11. Students have to attach plagiarism report in final spiral bound with duly signed by SGP guide.
12. Students have to bring internal review card hard copy on the day of internal review exam, after that they will bring filled review card on the day of external review.

D. Student Learning Outcome:

After the completion of the course students will able to

CO1	Identify a range of solutions, critically evaluate and justify proposed design solution.
CO2	Manage learning & self-development including development of organizational skills, time management, effective use of scientific literature and discriminating use of Web resources.
CO3	Apply a wide range of principles and tools available to the software developer such as choice of the algorithm, language, software libraries etc.
CO4	Write and test programs using appropriate test cases.
CO6	Evaluate system in terms of general quality attributes and possible trade-offs presented within the given problem/system.

Course Articulation Matrix:

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	2	-	2	2	2	2	-	1	3	3	2
CO2	3	3	1	2	1	2	2	2	2	-	1	3	3	2
CO3	3	1	3	3	3	1	2	2	3	-	2	3	3	2
CO4	3	1	1	3	1	-	-	1	2	-	2	2	2	2
CO5	3	-	-	-	-	-	2	3	3	3	3	2	2	1
CO6	3	2	1	2	1	-	-	1	2	-	1	1	3	1

The correlation levels 1, 2, or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, “-” is placed.

E. Recommended Study Material:

❖ Reference book:

1. John M Nicolas, Project Management for Business, Engineering and Technology, Elsevier.
2. Sanjay Mohapatra, Software Project Management, Cengage Learning
3. Clive L. Dym, Patrick Little, Elizabeth J. Orwin, “Engineering Design – A Project Based Introduction”, Wiley India Pvt. Ltd.
4. B. Hughes & M. Cotterell, “Software Project Management”, Tata McGraw Hills.

❖ Web Materials:

1. <https://status.net/templates/project-report/>
2. https://www.tutorialspoint.com/software_engineering/software_project_management.htm
3. <https://www.geeksforgeeks.org/coding-standards-and-guidelines/>
4. <https://www.altexsoft.com/blog/engineering/8-ways-to-improve-software-testing-through-planning-work-environment-automated-testing-and-reporting/>
5. <https://nptel.ac.in/courses/106/105/106105218/>
6. <https://www.youtube.com/watch?v=T3q6QcCQZQg>
7. <https://www.scribbr.com/category/research-paper/>

IT446: SUMMER INTERNSHIP-II

A. Credit and Hours:

Teaching Scheme	Project		Total	Credit
	Internal	External		
Hours	45	45	90	3
Marks	75	75	150	

B. Instructional Method and Pedagogy:

- Summer internship shall be at least 90 hours during the summer vacation only.
- Department/Institute will help students to find an appropriate company/industry/organization for the summer internship.
- The student must fill up and get approved a Summer Internship Acceptance form by the company and provide it to the Coordinator of the department within the specified deadline.
- Students shall commence the internship after the approval of the department Coordinator. Summer internships in research centers is also allowed.
- During the entire period of internship, the student shall obey the rules and regulations of the company/industry/organization and those of the University.
- Due to inevitable reasons, if the student will not able to attend the internship for few days with the permission of the supervisor, the department Coordinator should be informed via e-mail and these days should be compensated later.
- The student shall submit two documents to the coordinator for the evaluation of the summer internship:
 - Summer Internship Report
 - Summer Internship Assessment Form
- Upon the completion of summer internship, a hard copy of “Summer Internship Report” must be submitted through the presentation to the coordinator by the first day of the new term.
- The report must outline the experience and observations gained through practical internship, in accordance with the required content and the format described in this guideline. Each report will be evaluated by a faculty member of the department on a satisfactory/unsatisfactory basis at the beginning of the semester.
- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the summer internship.

C. Format of Summer Internship Report:

The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- DESCRIPTION OF THE COMPANY/INDUSTRY/ORGANISATION: Summarize the work type, administrative structure, number of employees (how many engineers, under which division, etc.), etc. Provide information regarding
 - Location and spread of the company
 - Number of employees, engineers, technicians, administrators in the company
 - Divisions of the company
 - Your group and division
 - Administrative tree (if available)
 - Main functions of the company
 - Customer profile and market share
- INTRODUCTION: In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.
- PROBLEM STATEMENT: What is the problem you are solving, and what are the reasons and causes of this problem.
- SOLUTION: In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.

- **CONCLUSIONS:** In the last section, summarize the summer internship activities. Present your observations, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
- **REFERENCES:** List any source you have used in the document including books, articles and web sites in a consistent format.
- **APPENDICES:** If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.

D. Learning Outcomes:

After completion of the course students, will able:

- To apply knowledge and skills learned in company/industry/organization to real- world problems.
- To solve engineering problems.
- To function in a team work.
- To work with teammates from other disciplines.
- To use experience related to professional and ethical issues in the work environment.
- To explain the impact of engineering solutions developed in a project, in a global, economic, environmental, and societal context.
- To finds relevant sources (e.g., library, Internet, experts) and gather information.
- To demonstrate knowledge of contemporary issues related with engineering in general.
- To use new tools and technologies

E. Student Learning Outcome:

At the end of the course, the students will be able to

CO1	Demonstrate ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
CO2	Cultivate an understanding of their multidisciplinary interest, including the skills, responsibilities and career path of professionals through practice-oriented and 'hands-on' working experience.
CO3	An exhibit foresight, independent thinking, resourcefulness, and the ability to make decisions.
CO4	Develop a right work attitude, self-confidence, interpersonal skills and ability to work as a team in an industry.

Course Articulation Matrix:

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	3	-	-	-	-	-	-	-	-	3	2
CO2	1	2	3	2	-	-	-	-	-	-	-	-	1	1
CO3	1	1	2	-	-	-	-	2	2	3	-	-	2	1
CO4	-	-	-	-	-	-	-	-	-	-	-	-	1	3

The correlation levels 1, 2, or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial

(High) If there is no correlation, “-” is placed.

B. Tech. (Information Technology) Programme

SYLLABI (Semester –8)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

IT461: MAJOR PROJECT

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	36	36	18
Marks	0	600(250+350)	600	

B. Outline of the Course:

- Software Project includes course work on a specialized Subject or a Seminar.
- The course work shall be related to the area of his/her project research work.
- Students have to take 3 months training to the other software industry as the project work.
- The major project work provides students an opportunity to do something on their own and under the supervision of internal guide as well as guide from industry.
- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Project will be evaluated at least thrice during the semester by internal guide of the project and final submission at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as solution of particular problem by applying principles of Software Engineering.
- A student has to produce some useful outcome by conducting experiments or project work.
- Student can learn all aspects & functionality of specialized software from the industry.
- Students have to submit SRS, SPMP, Design documents, Code and Test Cases in form of Project report.

C. Instructional Method and Pedagogy:

Following are the General guidelines:

- 1) Semester 8th, teaching scheme is Practical 36 hours, with 20 credits worth of 600 marks (Out of 600 marks, 250 marks of internal and 350 marks of external evaluation)

Note:

- a) Each defined project definition should be from Industry/Research organization/ Govt. organization / technical issues/Real world problems.
- b) If industry defined project then maximum 2 or 3 students are allowed per project group. If in-house project then no group is allowed.
- c) The students are required to identify their problem and they are required to follow all the rules and instructions issued by department.

Final Year Project Policy:

1. Process for NOC:

Following is the process for 8th semester project for definition and company approval:

1. Select your domain
2. Select your company
3. Approve company from HoD Sir/TPR
4. Issue recommendation letter from TPR (write company address in to, fill the details of students and bring its printed copy and submit to concern TPR.)
5. Issue confirmation letter from company with brief definition, tools & technology (submit Xerox copy to concern TPR)
6. Approve definition form HoD Sir / Sr. faculty/TPR
7. The Process for Approval of the Project Definition:
 8. The students must meet and discuss the definition of their final semester project with the HoD Sir/Faculty Member-Guide and get his approval by verifying to see that the following parameters:
 7. The proposed project quality should be up to the status of a B.Tech final semester project quality.
 8. The project should not be a conventional project.
 9. The project should not be a purchased/3rd party developed project.
 10. If the project is being carried forwarded from previous years then it must add substantial value to the previously done work on the project.
 11. The project should be novel, original and having a possibility of good impact if the proposed solution get implemented.
 12. Even if student claim it to be an Industry defined project, it should not be based on industry whose main objective is to make final semester project and give it to students.
 13. Issue NOC from TPO (submit Xerox copy to Concern TPR)
 14. In order to improve student's performance we are doing following exercise:
 1. Industrial visit
 2. Review and suggestions from internal guide
 3. Feedback from external guide

2. Process for Continuous Evaluation:

Following is the process for 8th semester project continuous evaluation:

1. Submit your project profile & synopsis to your internal Guide.
2. Report weekly to your internal guide with filled weekly report (At least 10 reporting is

mandatory).

3. 2 internal presentations & 1 final presentation with project demonstration are required. Each internal presentation carries 50 Marks, 100 marks for report and 50 marks from internal guide & External presentation carries 350 marks.
4. Observation Canvas: Observation points from survey, Users, Stockholders, Activities
5. Ideation Canvas: People, Activities, Problem (that you are going to solve), Situation/Context/Location, Possible Solutions
6. Project Development Canvas: Purpose, People, Product Experience, Product Functions, Product Features, Components, Customer Revalidation
7. Business Model Canvas: Applications, Usage & Outcome
8. Submit hard binding report with CD.

3. Continuous evaluation Marks:

Project guide has to put the marks according to grade.

Range is given below:

A+ : 47-50

A : 44-46

A- : 41 - 43

B+ : 36-40

B : 31- 35

B- :26 - 30

C+ : 21-25

C : 16 - 20

C- : <=15

As per the performance of students, guide can give the marks. For example: A+: One can give 47 - 50 as per performance.

D. Recommended Chapters/sections

1. Microscope Summery
2. Details of candidate and supervisor along with certificate of
 - original work,
 - Assistance, if any,
 - Credits,
3. Aims and Objectives
4. Approaches to Project and Time Frame

5. Project Design Description with appendices to cover Flow charts/Data Flow Diagram – Macro/Micro Level Source code, If any Hardware platform Software Tools Security Measures Quality Assurance Audit Ability Test Date and Result.

E. Student Learning Outcome:

At the end of the course, the students will be able to

CO1	Explore the new ideas & the possible areas to work ahead.
CO2	Use the various methodologies useful for doing project work.
CO3	Investigate the chosen topic in depth. This implies collecting and reviewing literature and understanding and interpreting the most up-to-date concepts and theories of your chosen academic field and/or project topic.
CO4	Apply the concepts and theories learnt in previous years of study and work placements

Course Articulation Matrix:

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	-	-	-	-	-	-	-	-	-	2	-
CO2	3	1	-	-	-	-	-	-	-	-	-	-	2	-
CO3	3	2	2	-	2	-	-	-	-	-	-	-	3	1
CO4	3	-	-	-	1	-	-	-	-	-	-	-	2	-

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1: Slight (Low) 2: Moderate (Medium) 3: Substantial

(High) If there is no correlation, “-” is placed.

F. Recommended Study Material:

❖ Web Link:

1. www.ieeeexplore.ieee.org
2. www.sciencedirect.com